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Medical device

Abstract

A medical device includes a treatment module configured to apply a treatment to a patient. The medical device includes an interface configured to operatively connect to a removable storage device storing authorization data that identifies a level of treatment authorization. The medical device includes a processing device configured to perform operations in response to receiving user input indicating a treatment should be initiated. The operations include determining whether the removable storage device is valid for use with the medical device. If the removable storage device is determined to be valid, the authorization data is accessed. The processing device determines whether the treatment is authorized based on the accessed authorization data. If the treatment is determined to be authorized, the treatment module is controlled to apply the treatment. If the treatment is determined to not be authorized, the treatment module is controlled such that the treatment is not applied.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/863,820, filed on Apr. 30, 2020, which is a continuation of U.S.

application Ser. No. 15/482,449, filed on Apr. 7, 2017, now U.S. Pat. No. 10,639,502, issued on May 5, 2020, which is a continuation of U.S. application Ser. No. 15/357,883, filed on Nov. 21, 2016, now U.S. Pat. No. 10,086,216, issued on Oct. 2, 2018, which is a continuation of U.S. application Ser. No. 13/271,870, filed on Oct. 12, 2011, now U.S. Pat. No. 9,526,920, issued on Dec. 27, 2016, which claims the benefit of U.S. Provisional Application No. 61/392,154, filed Oct. 12, 2010, U.S. Provisional Application No. 61/405,405, filed Oct. 21, 2010, U.S. Provisional Application No. 61/405,757, filed Oct. 22, 2010, and U.S. Provisional Application No. 61/483,445, filed May 6, 2011, each of which is incorporated herein in its entirety for all purposes.

FIELD

(1) This description relates to a medical device.

BACKGROUND

(2) Medical devices can provide treatment for a variety of health conditions. In some instances, a patient has a degree of control over treatment with a medical device. For example, a patient may be able to initiate treatment with a medical device. The capabilities of a medical device determine to a large degree the way that the patient and others interact with the medical device. In particular, it is important that a medical device be capable of providing effective treatment and a positive patient experience.

SUMMARY

(3) In one aspect, a medical device includes: a treatment module configured to apply a treatment to a patient; one or more processing devices configured to: disallow treatment using the treatment module until treatment authorization occurs; determine that treatment authorization occurs; and in response to determining that treatment authorization occurs, permit treatment to be applied using the treatment module.

(4) Implementations may include one or more of the following features. For example, treatment authorization occurs via payment. Treatment authorization occurs via patient identification. Treatment authorization is linked to geographic location. Treatment authorization occurs via authorization data stored on a removable medium.

(5) Implementations may include one or more of the following features. For example, the medical device includes a wireless communication module operable to receive the activation code over a wireless communication link. The wireless communication link is a Bluetooth or cellular communication link. The medical device is configured to receive the activation code from an enabling device configured to activate a limited number of medical devices and to not activate more than the limited number of medical devices. The one or more processing devices being configured to receive the activation code includes the one or more processing devices being configured to access the activation code from a removable medium. The activation code includes authorization information indicating a number of treatments authorized using the medical device.

(6) In another general aspect, an enabling device includes: a communication module operable to wirelessly transmit one or more activation codes to a medical device, the one or more activation codes being capable of activating the medical device; one or more data storage devices storing authorization information indicating a number of authorized activations of medical devices that the enabling device is authorized to perform; one or more processing devices configured to: receive user input indicating that a medical device should be activated; determine, based on the authorization information, whether activation of the medical device is authorized; if the determination indicates that the activation of the medical device is authorized: control the wireless communication module to transmit the one or more activation codes, and update the authorization information to decrease the number of authorized activations remaining for the enabling device; if the determination indicates that the treatment is not authorized, control the treatment module such that the one or more activation codes are not transmitted.

(7) Implementations may include one or more of the following features. For example, the enabling

device is associated with an inventory of medical devices, and the number of authorized activations of medical devices is based on the associated inventory. The communication module is further configured to receive, over a network, second authorization information altering the number of authorized activations the enabling device is authorized to perform, and the one or more processors are further configured to update the information indicating the number of authorized activations to alter the number of authorized activations by the additional number indicated by the second authorization information. The communication module is further configured to receive, over a network, second authorization information indicating an additional number of authorized activations the enabling device is authorized to perform, and the one or more processors are further configured to update the information indicating the number of authorized activations to increase the number of authorized activations by the additional number indicated by the second authorization information.

(8) In another general aspect, a medical device includes: at least one treatment module configured to apply a treatment to a patient, the treatment module including at least one ultrasound transducer and at least one driver circuit coupled to the ultrasound transducer; one or more data storage devices; an interface configured to operatively connect to a removable storage device storing authorization data that identifies a level of treatment authorization; and at least one processing device configured to perform the following in response to receiving user input indicating a treatment should be initiated: determine whether the removable storage device is valid for use with the medical device; if the removable storage device is determined to be valid, access the authorization data; determine whether the treatment is authorized based on the accessed authorization data; if the treatment is determined to be authorized, control the treatment module to apply the treatment; and if the treatment is determined to not be authorized, control the treatment module such that the treatment is not applied.

(9) Implementations of any of the aspects may include one or more of the following features. For example, to determine whether the removable storage device is valid for use with the medical device, the at least one processing device is configured to: access a serial number of the removable medium; and determine that the removable medium is valid if the serial number is within a predetermined range of values. The at least one processing device is configured to record compliance data for the treatment on the removable storage device if the treatment is applied. The compliance data indicates a time, date, and duration of the treatment. The treatment module includes at least one ultrasound transducer and at least one driver circuit coupled to the ultrasound transducer. To control the treatment module to apply the treatment, the processing device is configured to control the driver circuit such that the driver circuit causes the ultrasound transducer to produce ultrasound with therapeutic properties. To control the treatment module such that the treatment is not applied, the processing device is configured to control the driver circuit such that ultrasound with therapeutic properties is not produced. The driver circuitry includes a signal generator and an ultrasound transducer driver.

(10) The authorization data indicates a number of authorized treatments, and the at least one processing device is configured to decrease the number of authorized treatments indicated by the authorization data after a treatment is applied. The authorization data indicates an authorized amount of treatment time, and the at least one processing device is configured to decrease the amount of authorized treatment time indicated by the authorization data after a treatment is applied. The medical device includes a communication module. The at least one processing device is configured to: receive authorization data that indicates a level of treatment authorization through the communication module; store the received authorization data on the removable storage device or a second storage device of the medical device; and in response to receiving the user input indicating a treatment should be initiated, determine whether the treatment is authorized based on the received authorization data. The communication module is a wireless communication module. The wireless communication module is a cellular communication module.

(11) The authorization data is received from a server system configured to: determine that payment has been made for a number of treatments; determine that the payment is associated with the medical device; generate the authorization data; and transmit the authorization data to the medical device. The processing device is further configured to provide, if the treatment is determined to not be authorized, an indication to the patient that more treatments need to be purchased. The processing device is configured to receive new authorization data after providing the indication to the patient that more treatments need to be purchased, the new authorization data identifying a number of additional treatments for which payment has been received. The authorization data is encrypted, and the processing device is further configured to decrypt the authorization data. The medical device includes a second storage device. The second storage device stores a device identifier that uniquely identifies the medical device, the authorization data is encrypted, the at least one processing device is further configured to decrypt the authorization data; and the device identifier is used to decrypt the authorization data. The medical device includes a payment module configured to receive payment for a number of treatments. The payment module is configured to receive payment through a code, a credit card, or a SIM card. The at least one processing device is configured to: record on the removable storage device or an internal storage device information indicating occurrences of treatment applied by the medical device; and indicate, on a display of the medical device, days during which treatment was applied and days during which treatment was not applied. The medical device stores information about a treatment regimen for use of the medical device; and the at least one processing device is configured to indicate compliance with the treatment regimen by displaying a calendar that indicates the days during which treatment was applied and the days during which treatment was not applied.

(12) Implementations may include one or more of the following features. For example, the enabling device is configured to activate the medical device by transmitting an activation code over a wireless communication link between the enabling device and the medical device, and the medical device is configured to allow treatments using the treatment module in response to receiving the activation code.

(13) In another general aspect, a medical device includes: a treatment module configured to apply a treatment to a patient, the treatment module including at least one ultrasound transducer and at least one driver circuit coupled to the ultrasound transducer, a display, one or more data storage devices, and one or more processing devices. The one or more processing devices are configured to: receive user input that a treatment should be initiated using the treatment module; in response to receiving the user input, apply a treatment using the treatment module; record on the one or more data storage devices information indicating occurrences of treatment using the treatment module; and indicate on the display days during which treatment was applied and days during which treatment was not applied.

(14) Implementations of any of the aspects may include one or more of the following features. For example, the days during which treatment was applied and days during which treatment was not applied are indicated on a calendar. Days of the calendar during which one or more treatments are applied are marked with a check mark and days of the calendar during which one or more treatments were not applied are unchecked. The one or more storage devices store information about a treatment regimen for use of the medical device, and the one or more processing devices are configured to indicate compliance with the treatment regimen on the calendar. The one or more processing devices being configured to indicate on the display days during which treatment was applied and days in which treatment was not applied includes the one or more processing devices being configured to automatically indicate on the display, in response to the medical device being powered on, days during which treatment was applied and days during which treatment was not applied.

(15) In another general aspect, a medical device includes: a treatment module operable to apply a treatment to a patient, the treatment module including at least one ultrasound transducer and at least

one driver circuit coupled to the ultrasound transducer; a primary or rechargeable battery configured to supply power to the medical device; a color display configured to display information to a user of the medical device; a wireless communication module; one or more data storage devices; one or more processing devices configured to: disallow treatment using the treatment module until an activation code is received; receive an activation code received by the wireless communication module over a wireless communication link; in response to receiving the activation code, permit treatment to be applied using the treatment module; store on the one or more data storage devices authorization information indicating a number of authorized treatments remaining for the medical device; receive user input indicating a treatment should be initiated; determine whether the treatment is authorized; if the determination indicates that the treatment is authorized, control the treatment module to apply the treatment and update the authorization information to decrease the number of authorized treatments remaining for the medical device; if the determination indicates that the treatment is not authorized, control the treatment module such that the treatment is not applied; record, on the one or more storage devices, compliance information indicating dates during which treatment was performed using the treatment module; cause the wireless communication module to transmit the recorded compliance information over the wireless communication link; and display on the color display a calendar indicating days during which treatment was performed and days during which treatment was not performed.

(16) In another general aspect, a medical device includes: a treatment module operable to apply a treatment to a patient, the treatment module including at least one ultrasound transducer and at least one driver circuit coupled to the ultrasound transducer; a primary or rechargeable battery configured to supply power to the medical device; a color display configured to display information to a user of the medical device; one or more data storage devices including a removable memory card and a non-removable storage device; one or more processing devices configured to cause the medical device to: access from the removable memory card information indicating a number of authorized treatments for the medical device; based on the accessed information, store authorization information indicating the number of authorized treatments for the medical device on the non-removable storage; receive user input indicating a treatment should be initiated; determine, based on the authorization information, whether the treatment is authorized; if the determination indicates that the treatment is authorized, control the treatment module to apply the treatment and update the authorization information to decrease the number of authorized treatments remaining for the medical device; if the determination indicates that the treatment is not authorized, control the treatment module such that the treatment is not applied; record on the one or more storage devices compliance information indicating dates during which treatment was performed using the treatment module; automatically display on the color display, in response to the medical device being powered on, a calendar indicating days during which treatment was performed and days during which treatment was not performed; access, from a second removable memory card, second authorization information indicating a second number of authorized treatments using the medical device; and based on the second authorization information, increase the number of authorized treatments remaining for the medical device stored on the one or more storage devices.

(17) The medical device includes means for determining the geographic location of the medical device, and treatment is not allowed to commence if the determined geographic location is outside an authorized geographic location. The medical device includes means for determining user identity, and treatment is not allowed to commence if the determined user identity does not match an authorized identity. The treatment module is configured to produce a pulsed ultrasound signal having a frequency in the range of 1 MHz to 2 MHz, consisting of pulses generated at a rate in the range of 100 Hz to 10 KHz with each pulse having a duration in the range of 10 microseconds to 2,000 microseconds. The pulsed ultrasound signal has a power intensity of 100 milliwatts per square centimeter or less. The medical device is a hand-held device configured to accelerate bone healing.

(18) In another general aspect a computer-implemented method includes: receiving user input through a user interface of a medical device, the user input indicating a treatment should be administered by the medical device, and the medical device includes a treatment module configured to apply a treatment to a patient; determining whether a removable storage device coupled to the medical device is valid for use with the medical device; in response to determining that the removable storage device is valid for use with the medical device, accessing authorization data stored on the removable storage device; determining that the treatment is authorized based on the accessed authorization data; and controlling the treatment module to apply the treatment in response to determining that the treatment is authorized based on the accessed authorization data.

(19) Implementations of any of the above aspects may include one or more of the following features. For example, the activation code includes the authorization information. The wireless communication link is a Bluetooth or cellular communication link. The one or more storage devices include a removable memory card; and the one or more processing devices configured to: record the compliance information to the removable memory card; and access an activation code stored on the removable memory card or another memory card.

(20) The removable memory card and the second removable memory cards are Secure Digital memory cards. The one or more processing devices being configured to record on the one or more storage devices compliance information indicating dates during which treatment was performed using the treatment module includes the one or more processing devices being configured to record, on the removable memory card and on the non-removable storage, compliance information indicating dates during which treatment was performed using the treatment module. The one or more processing devices being configured to indicate on the display days during which treatment was performed and days in which treatment was not performed includes the one or more processing devices being configured to automatically indicate on the display, in response to the medical device being powered on, days during which treatment was performed and days in which treatment was not performed. The one or more processing devices are further configured to receive an authorization code indicating a number of treatments that are authorized for the medical device and update the authorization information such that the number of treatments indicated by authorization code are added. The one or more processing devices are configured to determine whether the treatment is authorized by determining if the authorization information indicates that at least one treatment is authorized for the medical device. The one or more processing devices are further configured to: access an identifier from the one or more storage devices; and cause the compliance information and the accessed identifier to be transmitted to a server system that is configured to receive the compliance information and the identifier.

(21) Implementations of any of the aspects may include one or more of the following features. For example, determining whether the removable storage device coupled to the medical device is valid for use with the medical device includes accessing a serial number of the removable medium and determining that the removable medium is valid if the serial number is within a predetermined range of values. Recording compliance data for the treatment on the removable storage device if the treatment is applied. The treatment module includes at least one ultrasound transducer and at least one driver circuit. Controlling the treatment module to apply the treatment includes controlling the driver circuit such that the driver circuit causes an ultrasound transducer to produce ultrasound with therapeutic properties. Controlling the treatment module such that the treatment is not applied includes controlling the driver circuit such that ultrasound with therapeutic properties is not produced. Receiving the authorization data, and storing the received authorization data on the removable device.

(22) Receiving the authorization data includes receiving the authorization data with a wireless communication module. The wireless communication module is a cellular communication module. The authorization data indicates a number of authorized treatments, and the computer-implemented method further includes decreasing the number of authorized treatments indicated by the

authorization data after a treatment is applied. The authorization data indicates an authorized amount of treatment time, and the computer-implemented method further includes decreasing the amount of authorized treatment time indicated by the authorization data after a treatment is applied. Storing a device identifier that uniquely identifies a medical device. The authorization data is encrypted, accessing the authorization code includes decrypting the authorization data, and the device identifier is used to decrypt the authorization data. Recording on the removable storage device or an internal storage device information indicating occurrences of treatment applied by the medical device. Indicating, on a display of the medical device, days during which treatment was applied and days during which treatment was not applied. Indicating compliance with a treatment regimen by displaying a calendar that indicates the days during which treatment was applied and the days during which treatment was not applied. Receiving payment for a number of treatments. Receiving payment for a number of treatments includes receiving payment at the medical device for a number of treatments, the payment being entered with a code, a credit card, or a SIM card.

(23) In another general aspect, a computer-implemented method includes: receiving user input that a treatment should be initiated using a treatment module of a medical device; in response to receiving the user input, controlling the treatment module to apply a treatment; recording on one or more data storage devices information indicating occurrences of treatment using the treatment module; and indicating, on a display of the medical device, days during which treatment was applied and days during which treatment was not applied.

(24) Implementations of any of the aspects may include one or more of the following features. For example, indicating days during which treatment was applied and days during which treatment was not applied includes indicating the days during which treatment was applied and the days during which treatment was not applied on a calendar. Indicating the days during which treatment was applied and the days during which treatment was not applied on a calendar includes marking days of the calendar during which treatment was applied with a check mark and displaying days of the calendar during which one or more treatments were not applied as unchecked. Storing information about a treatment regimen for use of the medical device, and indicating compliance with the treatment regimen on the calendar. Indicating the days during which treatment was applied and the days during which treatment was not applied occurs in response to the medical device being powered on.

(25) Determining the geographic location of the medical device, and disallowing commencement of treatment using the treatment module if the determined geographic location is outside an authorized geographic location. Determining user identity, and disallowing commencement of treatment using the treatment module if the determined user identity does not match an authorized identity. Controlling the treatment module to apply a treatment includes controlling the treatment module to produce a pulsed ultrasound signal having a frequency in the range of 1 MHz to 2 MHz, consisting of pulses generated at a rate in the range of 100 Hz to 10 KHz with each pulse having a duration in the range of 10 microseconds to 2,000 microseconds. The pulsed ultrasound signal has a power intensity of 100 milliwatts per square centimeter or less.

(26) According to another general aspect, a computer-readable storage medium storing instructions that, when executed by one or more processing devices, cause the one or more processing devices to perform the operations of the computer-implemented methods.

(27) According to another general aspect, a medical device includes: at least one treatment module configured to apply a treatment to a patient, the treatment module including at least one ultrasound transducer and at least one driver circuit coupled to the ultrasound transducer; an interface configured to operatively connect to a removable storage device storing authorization data that identifies a level of treatment authorization; and at least one processing device. The at least one processing device is configured to perform the following in response to receiving user input indicating a treatment should be initiated: determine whether the removable storage device is valid for use with the medical device; if the removable storage device is determined to be valid, access

the authorization data; determine whether the treatment is authorized based on the accessed authorization data; if the treatment is determined to be authorized, control the treatment module to apply the treatment by producing ultrasound with therapeutic properties; and if the treatment is determined to not be authorized, control the treatment module such that the treatment is not applied.

(28) The one or more processing devices are further configured to: identify a treatment regimen that identifies a prescribed use of the medical device; compare the information about a recorded use to the prescribed use of the medical device; and generate information indicating the degree that the recorded use matches the prescribed use.

(29) The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features and advantages will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a medical device.
- (2) FIG. 2 is a block diagram of the medical device.
- (3) FIG. 3 is a diagram of the medical device configured to display information.
- (4) FIG. 4 is a chart illustrating examples of information that may be displayed.
- (5) FIG. 5 is a flow diagram illustrating a process for displaying information.
- (6) FIGS. 6A to 6C are diagrams illustrating user interfaces for the medical device.
- (7) FIG. 7 is a diagram illustrating a medium for authorizing treatment using the medical device.
- (8) FIG. 8 is a diagram of a system for authorizing medical treatments.
- (9) FIGS. 9A and 9B are flow diagrams of a process for authorizing medical treatments.
- (10) FIG. 10 is a diagram illustrating a system for collecting information related to a medical device.
- (11) FIG. 11 is a flow diagram illustrating a process for sending information from a medical device.
- (12) FIG. 12 is a flow diagram illustrating a process for collecting information.
- (13) FIG. 13 is a diagram of a process for storing patient information.
- (14) FIG. 14 is a diagram of the medical device.
- (15) FIG. 15 is a diagram of a main operating unit of the medical device.
- (16) FIG. 16 shows authorized geographic areas.
- (17) FIG. 17 is a block diagram of geo-locking components of the medical device.
- (18) FIG. 18 is a block diagram of device authorization components of the medical device.
- (19) FIG. 19 is a flowchart depicted use of the medical device.
- (20) FIGS. 20 and 21 are diagrams illustrating a system for activating a medical device.

DETAILED DESCRIPTION

(21) In some implementations, a medical device can control its operation to apply treatments that have been purchased and to not apply treatments that have not been purchased. For instance, the medical device can be authorized to provide a particular number of treatments purchased for a patient. When the purchased treatments are exhausted, the medical device can be authorized to perform additional treatments based on additional payment. For example, a user can purchase a card or other removable medium that authorizes a number of additional, prepaid treatments with the medical device. The medical device can access authorization data stored on the removable medium, determine that treatment is authorized, and perform the appropriate treatment.

(22) In addition, or as an alternative, to the above-mentioned features, the medical device can also store information and provide information to a patient. For example, the medical device may provide instructions for using the medical device or information about a particular health condition.

The medical device can select the information to provide based a particular health condition of the patient or other information about the patient.

(23) In addition, or as an alternative, to the above-mentioned features, the medical device can record compliance information that indicates occurrences of treatments using the medical device. For example, the compliance information can indicate days and/or times that a patient has performed treatments using the medical device. The compliance information can also indicate the degree to which a patient has complied with a particular treatment regimen. The compliance information can be collected at a server system, and the information can be accessed by multiple parties. The level of access that a party receives can be limited based on the party's relationship to the patient. The server system can receive, store, and provide access to compliance information from multiple medical devices operated by different patients. The medical device can also record errors in the operation of the medical device, send the errors to a server system, and receive service information to address the errors.

(24) In addition, or as an alternative, to the above-mentioned features, the medical device can display information to a patient that indicates the recorded compliance information. The medical device can display a calendar indicating days on which treatment was performed and days on which treatment was not performed. The calendar can also indicate whether or not the treatments that were performed occurred according to the scheduled treatments indicated by a particular treatment regimen for the patient.

(25) In addition, or as an alternative, to the above-mentioned features, the medical device can limit restrict unauthorized use based on an identity of a user or the geographical location of the medical device. The medical device can determine the geographic location of the medical device and not allow treatment to commence if the determined geographic location is outside an authorized geographic region. The medical device can also determine the identity of a user and not allow treatment if the determined identity is not any authorized identity.

(26) Some implementations of the medical device may provide the following advantages. For example, rather than incur a large initial expense by purchasing a medical device with unlimited treatments, in some instances, a patient or third-party payer may pay for only the treatments that are prescribed to the patient. By purchasing treatments as needed, patients and third-party payers may also spread expenses over time. Also, when treatments can be authorized after an initial set of treatments is exhausted, the additional use of the medical device can be easily added. Additional treatments can generally be authorized without service, repair, or reconditioning of the medical device. In some instances, the patient may enter payment directly at the medical device and receive treatment with minimal delay after payment is complete. In addition, by providing only treatments that have been purchased, excessive treatment and misuse of the medical device can be deterred.

(27) Some implementations of the medical device may provide the following additional advantages. Information can be displayed to the patient during treatment, including, for example, information about a health condition of the patient, instructions for using the medical device, and information about the patient's compliance with a treatment regimen. Messages may, for example, encourage, motivate, inform, entertain, and instruct the patient.

(28) Some implementations of the medical device may provide the following additional advantages. Compliance information may be collected and sent to a server system. Security measures can be implemented so that insurance companies, physicians, and caretakers can receive access to the compliance information based on their relationship to the patient. The server system may receive information about errors that occur during operation of the medical device, and the server system may send service information to address the errors.

(29) Some implementations of the medical device may provide the following additional advantages. Medical devices can be maintained in a deactivated state until activated in a controlled process, reducing incentive for theft and misuse. The ability of devices to activate medical devices and authorize medical treatments can be controlled. The number of medical device activations that

can be performed by a device can be limited to the inventory associated with the device.

(30) Some implementations of the medical device may provide the following additional advantages. Information indicating a patient's usage of a medical device over time and compliance with a treatment regimen can be displayed. The patient's compliance with a treatment regimen can be easily discernable from a calendar display indicating days on which treatment occurred.

(31) Some implementations of the medical device can limit use by unauthorized users.

(32) Some implementations of the medical device can also limit use outside an authorized geographical area.

(33) Referring to FIG. 1, a patient is shown using a medical device **10** that includes a treatment module for applying a treatment to the patient. In the example illustrated, the medical device **10** is a portable ultrasonic treatment device. The treatment module may include, for example, one or more ultrasound transducers **16** and at least one driver circuit coupled to the ultrasound transducers **16**.

(34) The medical device **10** can include a control unit **12** that controls the operation of the transducers **16**. The control unit **12** can include the transducer driver circuit. The medical device **10** can also include cables **18** that can carry power, data, and control signals between the control unit **12** and the transducers **16**.

(35) The medical device **10** can include a placement module **14** that couples the transducers at a location of the patient's body where treatment is needed, for example, over a fractured bone or next to damaged connective tissue. The placement module **14** can include a band, sleeve, applicator, or other connector to fasten the one or more transducers to a treatment site. An ultrasound conducting gel **20** can be applied to the skin of the patient to enable the ultrasound to propagate effectively to the patient's tissue.

(36) The medical device **10** can use low intensity, high-frequency acoustic energy (ultrasound) to treat injuries, defects, or pathologies. For instance, the ultrasonic treatment device can be designed to treat injuries, defects, or pathologies of bones or connective tissue, and, in some instances, can increase cellular level activity that leads to healing of ischaemic or grafted tissue. The medical device **10** may be used as an adjunct to surgical repair, in order to speed healing, or in some cases can be used alone to heal tissue injuries without surgery (e.g., for degenerative diseases such as osteoarthritis, tendinosis, and tendonitis). The medical device **10** can be suitable for use in treatment of bone fractures and/or connective tissues associated with joints, such as those in the hand, foot, wrist, ankle, knee, elbow, hip, shoulder, back, and neck.

(37) For example, following surgery, the medical device **10** can be applied non-invasively to the outside of the body (e.g., coupled to the skin with coupling media, such as a gel) in the region of the repaired tissue. The medical device **10** can be operated to transmit ultrasound (for example, in the form of pulses) into the tissue in need of treatment, or at the interface with the uninjured tissues. Exposure to the ultrasound can stimulate a faster, better quality repair of the tissue. At a bone interface, the ultrasound can also stimulate bone repair and bone ingrowth into repair or graft tissue. This can give rise to a faster, stronger repair and improved integration of the interface between, for example, tendon, ligament, and bone. The ultrasonic treatment device may also be used to non-invasively treat pathologies of connective tissues, such as osteoarthritis, ligament and tendon conditions, without the need for a surgical procedure.

(38) Referring to FIG. 2, the control unit **12** of the medical device **10** can include a processing device **50** that executes instructions stored on a storage device **52**. The processing device **50** can include one or more processing devices. The storage device **52** can include one or more storage devices, one or more of which may be removable. The control unit **12** can also include a driver circuit **54**, a user interface module **60**, a payment module **62**, a communication module **64**, and a power supply **68**.

(39) By executing the instructions stored on the storage device **52**, the processing device **50** can, for example, determine whether a treatment is authorized. If treatment is authorized, the processing device **50** can control the treatment module (for example, driver circuit **54** and transducers **16**) to

apply the treatment. Applying the treatment can include controlling the driver circuit **54** to produce ultrasound with therapeutic properties. Controlling the driver circuit **54** to produce ultrasound can include activating the driver circuit **54**, for example, supplying power to the driver circuit **54**, sending control signals to the driver circuit **54**, or causing the driver circuit **54** to produce a particular output. If the treatment is not authorized, the processing device **50** can control the treatment module such that the treatment is not applied. For example, the processing device **50** can control the driver circuit **54** such that ultrasound with therapeutic properties is not produced. Controlling the driver circuit to not apply treatment can include not activating the driver circuit **54**, deactivating the driver circuit **54**, setting the output of the driver circuit **54** (for example, setting the amplitude to zero), and/or otherwise limiting or preventing treatment. The processing device **50** can also be configured to control other components described below, for example through instructions stored on the storage device **52**.

(40) The processing device **50** can determine whether a treatment is authorized by, for example, accessing authorization data. If, for example, accessed authorization data is invalid, is for a different medical device **10**, has expired or all treatments associated with the code have been expended, or if no authorization data can be accessed, the processing device **50** can determine that treatment is not authorized. On the other hand, if valid authorization data can be accessed, the processing device **50** determines whether at least one treatment using the medical device **10** is authorized. The authorization data may or may not include an authorization code that indicates that payment for treatments has occurred. The authorization data can be stored in the storage device **52**, for example, or, as described further below, in a removable medium.

(41) The storage device **52** can store a device identifier, such as a serial number, that identifies the particular medical device **10**. The device identifier can uniquely identify the medical device **10** and distinguish it from all other ultrasonic treatment devices, even those of the same type or model. The storage device **52** can also store information about the treatments that are authorized for the medical device **10**, for example, a number of treatments that are authorized or an authorization code that authorizes treatments.

(42) The driver circuit **54** can be configured to send drive signals that cause the transducers **16** to generate ultrasound with therapeutic properties. The driver circuit **54** can include a signal generator **56** that generates a signal and a transducer driver **58** that drives the transducers **16** according to the generated signal. In an implementation, the ultrasound generated by the transducers **16** can include low intensity ultrasound (for example, 100 mW/cm^{sup.2} or less) having a frequency ranging between about 1 and 2 MHz, more particularly about 1.5 MHz. The ultrasound can be pulsed, with a pulse width ranging from about 10 to 2,000 microseconds, more particularly about 200 microseconds, with a repetition frequency ranging from about 100 Hz to about 10 KHz, more particularly about 1 KHz.

(43) The user interface module **60** can provide information to the patient and enable treatment to be initiated. The user interface module **60** may include one or more input devices or controls, for example, buttons, a keypad, or a touch-sensitive screen. The user interface module **60** may be used by a patient or other person, for example, to enter user input that indicates that a treatment should be administered by the medical device. When the processing device **50** determines that treatment is not authorized, the processing device **50** can provide an indication to the patient on the user interface module **60** that more treatments need to be purchased.

(44) The user interface module **60** may also include one or more output devices, for example a screen, a liquid crystal display, or lights. For example, the interface module **60** can include a screen **72**, for example, a liquid crystal display (LCD), a thin-film transistor (TFT) display, a field sequential display, or an organic light-emitting diode (OLED) display. The interface module **60** can also include light-emitting diodes (LEDs) and other indicators. The interface module **60** may include a speaker or other device that can produce sound (not shown), or other output devices. The user interface module **60** may also include input capabilities or input devices (not shown), for

example, buttons, one or more keypads, and other controls. The screen **72** may be touch-sensitive to receive input from a user. The user interface module **60** can also include an interface to access a removable storage medium, such as a subscriber identity module (SIM) card, a Secure Digital (SD) card, or other types of removable storage media.

(45) The payment module **62** can enable a patient to enter payment at the control unit **12**, or to receive information indicating prior payment. Payment can be enabled through one or more methods. The payment module **62** can include a credit card reader that reads a card and charges treatment to a credit card, debit card, or similar card that is swiped at the control unit **12**. The payment module can include a SIM card reader, and a patient may purchase a SIM card that includes information that represents one or more payments made for treatment with the ultrasonic treatment device. The payment module can include a reader for reading other types of removable media, for example, a SD card or other flash memory device. The control unit **12** can be configured to receive payment in the form of a code or other user input that may be entered on the interface module **60**. Some implementations may exclude the payment module **62**. For instance, in some implementations, provisions may be made to allow payment remotely from the device **10**, for example, at a computer connected to a network.

(46) The communication module **64** can be configured to send payment information to a remote system and/or receive authorization information that authorizes additional treatments using the medical device **10**.

(47) In some implementations, the processing device **50** is configured to receive an authorization code or other authorization data through the communication module **64** and to store the received authorization code in the storage device **52**. The communication module **64** can enable communication with a server system, client system, or other computer system over a wired or wireless connection. The communication module **64** may enable a communication link that is wired or wireless. The communication module may enable communication over, for example, Ethernet, Universal Serial Bus, **502.11**, Bluetooth, Zigbee, cellular networks, and other communication links. In one implementation, the communication module **64** can include a cellular transceiver **66** to receive and/or transmit information over a cellular network. The communication module **64** may also enable communication through multiple communication links.

(48) The communication module **64** can be configured to send payment information to a remote system and/or receive authorization information that authorizes additional treatments using the medical device **10**. The processing device **50** can be configured to receive an authorization code through the communication module **64** and to store the received authorization code in the storage device **52**.

(49) A power supply **68** can provide power to the components of the medical device **10**, including the driver circuit **54**, the processing device **50**, the storage device **52**, the payment module **62**, the communication module **64**, and the user interface module **60**. The power supply **68** can include a battery that is integrated into the control unit **12** or is removable. The battery can be primary battery or a rechargeable battery, and the power supply **68** can include a detachable power adapter that can charge a rechargeable battery.

(50) When a user performs treatment using the medical device **10**, the medical device **10** can collect and store compliance information. Collecting compliance information can include recording information about use of the medical device **10**, for example recording the number of treatments that are performed. Compliance information can include a number of treatments provided by the medical device **10**, a date and time that a treatment was provided by the medical device **10**, and/or a duration that a treatment was provided by the medical device **10**. Information about multiple uses or treatments with the medical device **10** can be collected.

(51) A treatment regimen that identifies a prescribed use of the medical device **10** can be identified. For example, the treatment regimen may be entered on the device after the health condition has been diagnosed or after the medical device **10** has been prescribed to the patient. Information about

a treatment regimen may be entered on the medical device **10** or received from a network, which may include a cellular network. The information about the recorded use of the medical device **10** can be compared to the information about the prescribed use of the medical. Information indicating the degree that the recorded use matches the prescribed use can be generated.

(52) Compliance information can be stored on the storage device **52**, on a removable medium, or both the storage device **52** and a separate removable medium. The compliance information may, but is not required to, include one or more results of a comparison between the recorded use of the medical device **10** and the treatment regimen of the patient.

(53) Referring to FIG. **3**, the medical device **10** can display information to the patient. For example, the medical device **10** can display information that relates to the particular patient using the medical device **10**, for example, information about a health condition of the patient, a treatment regimen of the patient, or a physician of the patient. The information displayed on the medical device **10** can thus be personalized to the particular patient that receives the medical device **10** and a particular health condition of the patient. In some instances, the information displayed may be selected to instruct, encourage, or entertain the patient. In addition, the information can provide advertisements and personalize treatment using the name or brand of, for example a particular physician, hospital, or insurance company.

(54) The information displayed on the medical device **10** may be organized into a plurality of messages **304**. Messages **304** can include a variety of media, including text, images, video, and sound. Messages **304** can be stored on the storage device **52** of the control unit **12**. Some messages **304** may be entered onto medical device **10** during manufacturing. For example, an initial set of predetermined messages **304** may be loaded onto a storage device **52** before it is shipped. Messages **304** may also be entered at other times to supplement the initially loaded messages **304**, including before a medical device **10** is dispensed to a patient and after a patient begins use of the medical device **10**. Messages **304** may be received with, for example, the communication module **64** and may be stored on the storage device **52**.

(55) Messages **304** can include information related to specific health conditions. For example, some messages **304** may relate to treatment of broken bones of the foot, and others may relate to treatment of broken bones of the arm. The medical device **10** can store messages **304** that relate to a wide variety of health conditions. To ensure that the messages **304** displayed to the patient are useful, the processing device **50** can access information that identifies a health condition of the patient, which can be stored on the storage device **52**.

(56) Based on the identified health condition, the processing device **50** can select one or more messages out of the set of messages **304** that are stored on the storage device **52**. For example, if the processing device **50** determines that the patient has a broken foot, the processing device **50** can select one or more messages **304** related to broken bones of the foot and treatment of a broken foot. The selected messages **304** can be displayed to the patient on the screen **72**. In some implementations, the screen **72** may be part of the interface module **60**, while in others the **72** screen may be integrated into the control unit **12**.

(57) Selected messages **304** can be displayed to the patient during treatment. For example, while a treatment is applied, the medical device **304** can display information to instruct the patient about proper use of the medical device **10**. In many instances, a patient receives only minimal instruction about the proper use of the medical device **10** when the medical device **10** is dispensed to the patient. A patient may forget the proper use of medical device and the details of a treatment regimen, especially when the medical device **10** is new. By providing messages **304** that instruct the patient how to use the medical device **10**, the patient may be more likely to perform treatment correctly. The instructive messages **304** can be selected based on the health condition of the patient and the associated treatment regimen for the health condition.

(58) The medical device **10** can select and display a variety of other messages **304** during treatment. For example, messages **304** can also provide general health information, such as,

“smoking inhibits bone healing” or “tell your doctor if you use blood thinners.”

(59) Messages **304** can also be selected based on a patient's compliance to a treatment regimen. The medical device **10** can store information that indicates when the patient should receive treatment. The medical device **10** can also record information indicating when treatment is actually performed. The medical device **10** can compare the planned or prescribed use of the medical device with the actual use of the device and determine how well the patient has complied with the prescribed treatment regimen. The medical device **10** can display messages **304** to the patient that directly or indirectly provide compliance information. For example, messages can provide direct feedback about a patient's compliance. Messages **304** can also be displayed that motivate, encourage, and remind the patient to follow a consistent treatment schedule. Messages **304** can also describe the benefits of continuing treatment or provide information about how the medical device **10** operates.

(60) Messages **304** can also provide physicians and others an opportunity to provide a personalized message. For example, one or more messages **304** may include the name of a patient's physician, the name of the patient's insurance company, or the logo for a hospital. Customized messages **304** can enable physicians and organizations to reinforce their brands and enhance the patient's experience during treatment. Messages **304** can also include contact information, for example, the phone number for the patient's primary physician. Messages **304** can include advertisements and paid content.

(61) Messages **304** can also be provided to entertain a patient during treatment and thus encourage the patient to complete the treatment. In some implementations, the medical device may enable the patient to acquire or input additional content to display on the medical device.

(62) Referring to FIG. **4**, a diagram illustrates the selection and display of messages **304** on a screen **72** of the medical device **10**.

(63) The processing device **50** of the medical device **10** can access information identifying a health condition of a patient. As illustrated, a health condition record **401** indicates that the health condition of the patient is a broken clavicle. Based on the identified health condition, the processing device **50** can select one or more messages from a plurality of messages **304**. The plurality of messages **304** may include a set **402** of predetermined messages **304**. Each message **304** in the set **402** may be associated with an identifier, as represented by the numbers from one to ten. From the set **402**, a subset **403** of messages **304** may be selected. The selected messages **304** can include messages **304** that relate to the particular health condition of the patient. The selected messages **304** can be ordered into a sequence **404** for display on the medical device **10**.

(64) The sequence **404** of messages **304** may be displayed on the screen **72** of the medical device **10**. In one implementation, the sequence **404** of messages **304** may begin to be displayed when treatment begins, and the sequence **404** may end roughly when treatment ends. In addition to the messages **304**, other information can be included, for example, information that describes the treatment being performed. For example, notifications **406** that indicate the time remaining until treatment is completed may be interspersed between other messages **304**.

(65) The messages **304** selected and the sequence **404** of the selected messages **304** can vary according to the needs of the patient and to limit unnecessary repetition. For example, instructions about how to use the medical device **10** may be selected and displayed for an initial set of treatments using the medical device **10**, but instructions may be omitted after many treatments have successfully been performed.

(66) Referring to FIG. **5**, an example of a process **500** for providing information is illustrated. The processing device **50** of the medical device **10** can be configured to perform the process **500**, for example, by executing instructions stored on the storage device **52**.

(67) A plurality of messages is stored on the medical device (**502**). For example, the storage device may store the plurality of messages. Messages may be entered on the storage device by a manufacturer of the medical device before the medical device is sold. Messages can also be entered

on the storage device by sales representatives, physicians, and others at other times.

(68) The medical device receives information about a health condition of a patient (504).

(69) For example, after a patient is diagnosed with a particular health condition, the health condition can be entered on the medical device. A physician, assistant, sales representative, or other person may enter information that indicates the health condition of the patient on the medical device. In addition, or alternately, the medical device may receive information about a health condition of the patient through the communication module. For example, the medical device may receive information about a prescription or diagnosis automatically over a network, without requiring any manual input. The medical device can store the information about the patient's health condition for later access. The health condition can be, for example, a health condition that is treatable by the medical device.

(70) The medical device selects messages for the patient (506). For example, the messages can be selected from the stored messages stored on the storage device. One or more messages can be selected based on the identified health condition of the patient. For example, if the patient has a broken ankle, messages can be selected that describe treatment of a broken ankle. The selected messages can include messages related to multiple health conditions. The selected messages can include instructions for using the medical device. The selected messages can include one or more messages that include information about a doctor that treated the patient or a medical office where the patient was treated. The selected messages can include one or more messages about the medical device or information about the provider of the medical device. The selected messages can include advertisements. The selected messages can include image data or video data.

(71) In some implementations, the medical device can store records indicating use of the medical device. For example, the medical device can record the number of treatments that have been performed using the medical device, the date and time that each treatment is performed, and/or the duration of each treatment. The information in these records, referred to generally as compliance information, indicates the manner in which treatments were performed using the device, from which a patient's compliance with a particular treatment regimen can be determined.

(72) Compliance information can be stored on the one or more storage devices 52. For example, the compliance information can be stored on internal memory of the medical device 10 and can also be stored on a removable medium, such as an SD memory card. Recording the compliance information on internal memory and the removable medium provides a backup in case one of the storage devices should fail. Additionally, the removable medium may be removed and used to transfer compliance information to other systems.

(73) The medical device can also identify a treatment regimen that corresponds to the health condition. For example, the medical device may receive the information from a prescription, a treatment regimen that is entered directly on the medical device, or the medical device may store a number of treatment regimens on the storage device. The medical device can access the records indicating use of the device. The medical device can compare the records indicating use of the medical device to the treatment regimen identified for the health condition of the patient.

(74) The medical device can provide an indication of compliance with the treatment regimen. For example, the medical device may provide an indication of compliance in one or more messages that are selected to be later displayed to the patient. The selected messages can also encourage compliance to a treatment regimen, for example, by praising the patient for past compliance or assuring the patient that continued treatment will bring good results. The selection of messages, including the selection of messages about compliance to the treatment regimen, can be based on the number of uses of the medical device indicated in the records that indicate use of the medical device.

(75) The medical device can also identify the language of the patient and select one or more messages in the language of the patient. The plurality of messages stored on the medical device can include messages in at least two languages. For example, some or all of the stored messages can be

included in multiple languages. The medical device can identify the language of the user, for example, based on user input, messages input on the medical device, information received by the communication module, or other information.

(76) The medical device can begin treatment (**508**). For example, a patient may enter input indicating that treatment should begin, and the medical device may control a driver circuit to drive an ultrasound transducer so that the ultrasound transducer produces ultrasound with therapeutic properties. The medical device can store and update records indicating use of the medical device.

(77) The medical device can display the selected messages (**510**). The selected messages can be displayed during treatment, for example, while the ultrasound with therapeutic properties is applied to the patient. The messages can be displayed on a liquid crystal display or other screen.

(78) Referring to FIG. **6A**, the medical device **10** can display information about a patient's compliance with a treatment regimen on a user interface **600a**. The user interface **600a** can be displayed on the screen **72** of the user interface module **60**. The user interface **600a** includes a calendar view **602** that indicates whether treatment was performed each day of, for example, the current month, or the current month and previous months. In the calendar space corresponding to each day that treatment was performed, a compliance indicator **604** can be displayed, for example, a colored square, a check mark, or other image or icon. In the space corresponding to each day in which treatment was not performed, a noncompliance indicator **606** can be displayed, for example, a different image or icon, such as a blank square or a red "X." Thus the user interface **600a** can visually distinguish the days during which treatment was performed from days during which treatment was not performed, providing an easily-understandable indication of recent compliance with the treatment regimen.

(79) In some implementations, information about the particular treatment regimen prescribed for the user of the medical device **10** is stored on the medical device **10**, and the compliance indicator **604** is displayed to indicate that a treatment performed on a particular day complies with the particular treatment regimen prescribed. In other words, rather than assuming that the treatment regimen requires one treatment each day, the medical device **10** compares times that treatments were performed to times that treatments were scheduled to be performed, as dictated by a treatment regimen. If a treatment regimen involves treatment every other day, for example, a neutral indicator can be displayed to represent days in which treatment was not scheduled and was not performed. The neutral indicator may be, for example, the day of the month that the day occurs. The noncompliance indicator **606** may be displayed, for example, only when treatment was scheduled to be performed on a day and treatment did not occur on that day. If treatment was performed on a day that treatment was not scheduled, an improper treatment indicator different from the noncompliance indicator **606** may be displayed for that day, distinguishing noncompliance by omitted treatment from noncompliance by performance of an unscheduled treatment. As a result, compliance relative to a treatment regimen can be accurately indicated when scheduled treatments are not scheduled every day.

(80) Similarly, compliance can be indicated for treatment regimens that dictate treatment multiple times in a day. For example, multiple compliance indicators **604** or multiple noncompliance indicators **606** can be displayed to indicate each treatment that was completed or missed that day.

(81) In other implementations, the medical device **10** displays the compliance indicator **604** for days that treatment was performed and displays the noncompliance indicator **606** for days that treatments were not performed, without regard to times that treatments were dictated by a prescribed treatment regimen. Thus even when the medical device **10** does not have access to information indicating a treatment regimen, the calendar view **602** indicates when treatments were performed, permitting the user or others to determine compliance with an appropriate treatment regimen.

(82) The user interface **600a** may display patient compliance for time period longer or shorter than a month, and for previous periods of time rather than, for example, the most recent weeks or

months.

(83) The medical device **10** can automatically display the calendar view **602** as the medical device **10** is powered on or at other times. For example, each time the medical device **10** is powered on, while the medical device **10** is initializing and for a period of time afterward, the calendar view **602** showing compliance can be displayed. The calendar view **602** can also be displayed to physicians, caretakers, and others. The calendar view **602** can be displayed automatically after particular functions of the medical device **10** are accessed, or in response to a request that the calendar view **602** be displayed.

(84) The medical device **10** can automatically display a total compliance to-date indication. For example, if ten days have elapsed since the start of a daily treatment regimen and the patient only used the device for eight out of the ten days, then the total compliance indicator can display 8/10 or 80% to indicate the overall level of compliance.

(85) The user interface **600a** can also display notification icons **610**. The notification icons **610** can vary in appearance according to the current status of the medical device **10**. The notification icons **610** can indicate, for example, the status and availability of communication links such as wireless connections, whether service is needed, that error or notification messages are available, the types or quality of connections with various modules, the remaining battery charge of the medical device, and other notifications.

(86) Referring to FIG. 6B, after the calendar view **602** is displayed, or after receiving user input, the medical device **10** can display a treatment timer **612** on a user interface **600b**. The treatment timer **612** can indicate the time remaining before a treatment is completed. For example, for a twenty-minute treatment, the treatment timer **612** can initially indicate the duration of treatment, twenty minutes. While a treatment is in progress, the treatment timer **612** can count down toward zero, reaching zero when the treatment ends. The notification icons **610** can also be displayed.

(87) Referring to FIG. 6C, an alternative user interface **620** includes a calendar view **622** indicating daily compliance with the treatment regimen and a treatment timer **624**. Days for which treatment was performed as indicated in the treatment regimen are indicated with a first marking **626**, while days for which planned treatment failed to be performed are indicated with a different marking **628**. Days in the future can be marked with their corresponding calendar numbers.

(88) Referring to FIG. 7, a removable medium **710**, for example, an SD card, USB device, or other removable memory device, can be used to authorize use of the medical device **10**. The removable medium **710** can store authorization data **720** that indicates a level of treatment authorization, for example, a number of treatments authorized or an amount of treatment time authorized using the medical device **10**.

(89) The medical device **10** can include an interface **712** that operatively connects to the removable medium **710**, permitting the processing device **50** to access the authorization data **720**. The interface **712** can include a slot that receives the removable medium **710** within the medical device **10**. The slot can be accessible to a user, permitting the user to replace the removable medium **710** with a different removable medium. The control unit **12** of the medical device **10** can define the slot and can include a cover that covers the slot.

(90) To obtain treatment authorization, the patient can obtain the removable medium **710**, which can be a prepaid medium that represents that payment has been made by or for the user. Removable media can store differing levels of treatment authorization. Different removable media may be sold with authorization data **720** that permits, for example, 50, 25, or 10 treatments. Treatment authorization may additionally or alternatively be indicated as an amount of time, for example, 1000, 500, or 100 minutes of treatment. In some implementations, the removable medium **710** can be purchased from a retail store or a physician's office. The fact that the patient obtained the removable medium **710** indicates that payment was made, and no additional verification of payment may be necessary to use the medical device **10**.

(91) The removable medium **710** may be a secure mode of communicating that a particular number

of treatments are authorized. The removable medium can include a copy-protection or anti-counterfeiting feature that can be used to determine whether the removable medium is genuine. For example, the removable medium can store an encoded value in a manner that the value is not easily duplicated or copied from one removable medium to another. The encoded value can be hardware-encoded or factory-set with a physical setting such that similar removable media cannot be altered to mimic the encoded value. In some implementations, the encoded value is a serial number that is embedded in non-writable storage of the removable medium. Each valid removable medium can have a unique serial number. Only removable media that have a serial number within a predetermined range of values can be considered genuine.

(92) In use, a user interacts with the medical device **10** to indicate that treatment should be initiated, for example, by pressing a button or entering other input. In response, the processing device **50** determines whether a removable medium is present. If no removable medium is present, the processing device **50** disallows treatment.

(93) If a removable medium such as the removable medium **710** is present, the processing device **50** determines whether the removable medium **710** is valid for use with the medical device **10**. For example, the processing device **50** determines whether a serial number or other value encoded in the removable medium meets predetermined criteria. In some implementations, the processing device **50** determines whether the value is within a predetermined set or range of values. The serial number can be a value that is not modifiable by a user, for example, a value that is fixed in the hardware configuration of the removable medium and cannot be copied onto a similar removable medium. Thus the processing device **50** can verify that the physical medium is valid. If a removable medium is not genuine, or is not compatible with or intended for the medical device **10**, the processing device **50** disallows treatment.

(94) If the removable medium **710** is genuine, the processing device **50** accesses authentication data **720** stored on the removable medium **710**. The authentication data **720** can indicate a number of treatments authorized or a number of treatment minutes that treatment is authorized. For example, when each treatment has a duration of twenty minutes, the authorization data **720** may indicate that ten treatments are authorized, or may indicate that two hundred minutes of treatments are authorized. If the authorization data **720** indicates that at least one treatment is authorized, or that one or more treatment minutes are authorized, the processing device **50** controls the treatment module to provide ultrasound with therapeutic properties. If the authorization data **720** indicates that no treatments are authorized, the processing device **50** disallows treatment.

(95) After the medical device **10** applies a treatment, the processing device **50** alters the authorization data **720** to indicate an updated level of authorization. For example, the medical device **10** can decrease the number of authorized treatments or decrease the number of authorized treatment minutes remaining. Modified authorization data that indicates an updated level of authorization can be stored on the removable medium **710**, for example, by overwriting the authorization data **720** that was stored before treatment began.

(96) The medical device **10** can also store compliance data **730** on the removable medium **710**. When the medical device **10** applies a treatment, the processing device **50** can store information about the treatment performed. For example, the compliance data **730** can indicate the time, date, and duration of the treatment applied, along with other treatment information. The removable medium **710** can thus include a compliance log that indicates use of the medical device **10** over time. The compliance data **730** can also indicate, for example, the degree that the use of the medical device **10** corresponds to planned or prescribed use of the medical device **10**. For example, the compliance data **730** can indicate days or times at which treatment was scheduled and whether treatment occurred at those days or times. Compliance data **730** can additionally or alternatively be stored on an internal storage device of the medical device, such as the storage device **52**.

(97) In some implementations, the authentication data **720** is encrypted, which can discourage tampering. In such implementations, the processing device **50** decrypts the authentication data **720**

before determining whether treatment is authorized. Also, after modifying the authentication data to indicate a decreased level of authorization, the processing device **50** encrypts the modified data and stores the encrypted data on the removable medium **710**.

(98) When additional treatments are desired, for example, after the treatment authorization of the authorization data **720** is depleted, a user can obtain a different removable medium that includes authorization data for additional treatments.

(99) In some implementations, the authorization data **720** directly authorizes the treatments, without the medical device **10** needing additional information or confirmation from another system. In some implementations, as described below, the medical device **10** verifies the authenticity of authorization data **720** by communicating with a server system or other device.

(100) Referring to FIG. **8**, a system **800** for authorizing medical treatments includes the medical device **10** connected to a server system **802** via a network **806**. The system **800** can also include a third-party server system **804** and a cellular network **808**. After payment is made by or for a patient, authorization data can be entered at or received by the medical device **10**. The authorization data indicates a level of treatment authorization, and can include an authorization code.

(101) A patient may desire to authorize medical treatments using the ultrasonic treatment device **10**. For example, the patient may receive the ultrasonic treatment device **10** in a condition in which treatments have not yet been authorized. As another example, the patient may have used treatments authorized for the ultrasonic treatment device **10** so that an insufficient number of treatments are currently authorized to complete treatment.

(102) To purchase additional treatments of the medical device **10**, the patient may provide payment information **810** at the medical device **10**. Payment information **810** includes payment entered or authorized at the medical device **10** and also information that indicates that payment has been made in another manner. For example, the patient may enter a credit card, a debit card, or another payment device into an appropriate reader of the medical device **10** and authorize a charge to an account owned by the patient. The patient may also enter an account number on the user interface module **60** of the medical device **10** to authorize payment. The patient may also purchase a prepaid medium, for example, a SIM card, a Secure Digital (SD) card, or a prepaid card with a magnetic strip, an optical code, or a printed code, from a store or physician's office. In one implementation, the prepaid medium may be a secure mode of communicating an amount of payment that has been paid. The patient may enter the prepaid medium at the ultrasonic treatment device **10** to indicate that payment has been made. The patient may also purchase treatments in a store or through a web site, may receive a confirmation code for the transaction, and may enter the confirmation code at the ultrasonic treatment device **10**. The system **800** can also be used to verify the validity of an authorization code received from a prepaid medium.

(103) The medical device **10** can send the payment information **810** to a server system **802**. The payment information **810** can be associated with a device identifier **812** that identifies the medical device **10**, which can also be sent to the server system **802**. In addition to, or instead of, sending a device identifier **812**, the medical device **10** may send a patient identifier that identifies the patient, together with the payment information. As will be described in greater detail below, the server system **802** can send an authorization code to the medical device **10** after receiving the payment information **810** and the device identifier **812**.

(104) The medical device **10** may send the payment information **810** and the device identifier **812** to the server system **802** through the network **806**. Alternatively, or additionally, the medical device **10** may initiate a communication using the cellular network **808** to send the payment information **810** and the device identifier **812** to the server system **802**. Payment information **810** can also be received through a removable medium, token, code or other indication that treatment is authorized.

(105) In one implementation, a prepaid medium can store an authorization code that can directly authorize treatments, so that the ultrasonic treatment device **10** is not required to transmit any information to the server system **802**. A prepaid medium can include an authorization code that can

enable treatments of the ultrasonic treatment device **10**, independent of a server system **802**. For example, a patient may purchase a SIM card or other device that stores an authorization code compatible with the ultrasonic treatment device **10**. The SIM card containing the authorization code can be entered at the ultrasonic treatment device **10** and the treatments authorized by the authorization code can be enabled on the ultrasonic treatment device **10**. Payment is received by the retail store or physician's office where the prepaid medium was obtained. The fact that the patient obtained the prepaid medium proves that payment was made, and no additional verification may be necessary. In some implementations, the ultrasonic treatment device **10** may verify that the authentication code included with the prepaid medium is authentic, and may ensure that the authentication code of the prepaid medium is not used multiple times (for example, by altering the data on the prepaid medium). In another implementation, a mechanical device or key may also be used to indicate authorization of additional treatments of the ultrasonic treatment device **10**.

(106) Treatments can also be purchased for a patient by a third-party payer, for example, an insurance company. A third-party server system **804** can transfer payment information **816** to the server system **802** with a patient identifier **818** that identifies the patient. The payment information **816** can include the information that completes the act of payment or indicates that payment has been made. The patient identifier **818** can include a name, prescription number, insurance policy number, or other identifier.

(107) The server system **802** can receive the payment information **810** and the device identifier **812** from the medical device **10**, or can receive the payment information **816** and the patient identifier **818** from the third-party server system **804**.

(108) The server system **802** can determine that payment has been made for a number of treatments, for example, using the payment information **810**, **816**. The server system **802** can also determine that patient is associated with the medical device, for example, using the patient identifier **818** or device identifier **812** associated with the payment information **810**, **816**. The server system **802** can use the received information and stored records to associate the payment with the patient to be treated with the medical device **10**. The server system **802** may store records that associate patient identifiers **818** and device identifiers **812** with particular patients and medical devices **10** so that payment can be applied for the correct patient and medical device **10**.

(109) The server system **802** can also generate an authorization code **814** that enables the medical device **10** to provide a number of treatments. The number of treatments authorized can be based on the amount of payment received. The server system **802** can send the authorization code **814** to the medical device **10** through the network **806** and/or through the cellular network **808**.

(110) The authorization code **814** can be encrypted or encoded so that the authorization code **814** enables treatments only for the particular medical device **10** associated with a particular payment received. In one implementation, the authorization code **814** can be encrypted so that the unique device identifier **812** or another unique decryption key is necessary to decrypt or decode the authorization code **814**. For example, the authorization code **814** can be encrypted using a symmetric-key or asymmetric-key encryption scheme.

(111) Using a symmetric-key or shared-key encryption system, a key may be used as both the encryption and decryption key. The key can be stored on both the server system **802** and the medical device **10**, for example, by the manufacturer of the medical device **10**. To prevent interception, the key may not be transmitted. The medical device **10** can send a patient identifier or a device identifier **812** unrelated to the key to identify the medical device **10**. The server system **802** can send the encrypted data to the medical device **10**, which can decrypt the data with the stored key.

(112) Using an asymmetric-key cryptography system, for example, a public key private key pair encryption system, the server system **802** can store an encryption key and the medical device **10** can store a corresponding decryption key. The server system **802** may encrypt the authorization code **814** using the encryption key and send the encrypted data to the medical device **10**. The

medical device **10** can include a stored decryption key that can decrypt the encrypted data. The decryption key can include the device identifier **812** or another key. In an implementation, the encryption key that encrypts messages for a particular medical device **10** may be known only to the server system **802**.

(113) Because the server system **802** can store records associating patients, medical devices **10**, and corresponding encryption keys, the system **800** may not require that the decryption key be sent to the server system **802**. If the device identifier **812** is used to decrypt an authorization code **814**, instead of sending the device identifier **812**, the medical device **10** can send another identifier, such as a patient identifier or a device identifier unrelated to the encryption scheme. The device identifier **812** can be independent of the encryption scheme so that interception of the device identifier does not compromise the encryption scheme.

(114) The medical device **10** can receive the encrypted authorization code **814** through the network **806** or the cellular network **808** and can decrypt the authorization code **814**. The medical device **10** can use the authorization code **814** to authorize a number of treatments of the medical device **10**. The authorization code **814** or information determined based on the authorization code **814** can be stored to indicate the number of treatments authorized. When the patient attempts to initiate treatment with the medical device **10**, the processing device of the medical device **10** can determine that authorized treatments remain for the medical device **10** and initiate treatment.

(115) The medical device **10** may also use the authorization code **814** to determine a change in treatment. For example, an authorization code **814** may indicate that treatment should be disallowed after a particular period of time has elapsed or if the patient does not apply a treatment for a period of time. The authorization code **814** may indicate that the number of treatments that are available each day should be changed, for example, from one treatment each day to two treatments each day. The authorization code **814** may indicate that the intensity of ultrasound produced by the medical device **10** should be changed, for example, that the intensity should be reduced if the patient is healing well.

(116) Referring to FIG. **9A**, a process **900** for authorizing medical treatments can include actions by a medical device, a server system, and/or a third-party system. The medical device can be an ultrasonic treatment device as described above.

(117) As illustrated, the process **900** can include payment for treatment by a patient at the medical device, payment by a third-party at a remote system, or both sources of payment. The actions performed by the medical device can be performed by one or more processing devices of the medical device configured to perform those actions. The server system can include one or more processing devices and one or more storage devices that store instructions that, when executed by the one or more processing devices, cause the processing devices to perform the various functions of the server system described below.

(118) A patient can attempt to initiate treatment with the medical device (**902**). For example, the medical device can receive user input indicating that a treatment should be administered by the medical device.

(119) The medical device can determine whether treatment is authorized (**903**). For example, the medical device can determine whether at least one treatment is authorized based on stored information that indicates the number of treatments that are authorized. The medical device can access an authorization code that has been received, for example, and determine whether treatment is authorized based on the accessed authorization code. The determination whether the attempted treatment is authorized can be performed in response to the attempt to initiate treatment in (**902**).

(120) If the determination indicates that treatment by the medical device is authorized, the medical device can control a treatment module to apply the treatment that the patient attempted. If the determination indicates that treatment is not authorized, the medical device can control the treatment module so that the treatment attempted by the patient is not applied. No treatment may be authorized for a medical device if, for example, all of the previously authorized treatments have

already been used or if the medical device has not received an initial authorization code to enable treatments.

(121) When treatment is not authorized for the medical device, the medical device can notify the patient that payment is needed to purchase additional treatments **(904)**. The example illustrated in FIG. 9A shows a scenario in which the medical device is not initially authorized to perform a treatment attempted by a patient, so additional payment and authorization of the medical device is needed.

(122) Payment can be received at the medical device **(906)**. The patient can then enter payment in one or more ways, including entering payment at the medical device using, for example, a credit card or a debit card to purchase additional treatments. The patient may also complete payment at a location other than the medical device, and enter proof of payment at the medical device. The patient may purchase treatments for example, at a store, at a medical office, or over the Internet. The patient may then enter proof of payment at the medical device in the form of, for example, a computer file, a code, or a SIM card.

(123) The medical device can then send payment information for the payment received and an identifier to the server system **(908)**. The identifier may be a device identifier that uniquely identifies the particular medical device used by the patient. In other words, the device identifier can identify not merely a model or type of medical device, but a single, particular medical device. The identifier may be a patient identifier that identifies a particular patient associated with the medical device.

(124) Treatments can also be purchased for a patient by a third-party, for example, an insurance company. The third-party system can receive, for example, a prescription for treatment of the patient using the medical device **(910)**. The third-party system can authorize one or more treatments using the medical device **(912)**. For example, the third-party system can authorize the treatments identified in the received prescription.

(125) The third party system can send payment information and an identifier to the server system **(914)**. The payment information can include information that enables a transaction to occur, for example, an authorization to charge an account or otherwise cause funds to be transferred, and can include information that indicates that payment has been performed. The identifier can identify the patient associated with the prescription that was received in action **(910)**. For example, the identifier can include a name of the patient, an insurance policy number for the patient, a prescription identifier, or other information relating to the patient. The identifier may also identify the medical device for the patient.

(126) The server system can receive payment information and an associated identifier from either the medical device or the third party system **(916)**. The associated identifier can be a device identifier that uniquely identifies the medical device. The server system can match the payment described in the payment information with the patient and the medical device of the patient **(918)**. The server system can store one or more associations between a patient and the medical device configured to apply a medical treatment to the patient. For example, the server system can store records that associate patients with particular medical devices, patient identifiers, and medical device identifiers.

(127) The server system may use one or more received identifiers to determine which patient and device are associated with a payment. Specifically, the server system can determine that payment has been made for the patient for a particular number of treatments by the medical device. The determination can be made based on the received information that payment has been made for the patient. The server system can identify the medical device associated with the patient based on the stored association between the patient and the medical device and, for example, based on a received device identifier that uniquely identifies the medical device. The server system may also record the determination that payment has been made for the user and the identification of the medical device associated with the patient.

(128) The server system can generate an authorization code that can authorize the medical device associated with the patient to perform the purchased treatments (920). The authorization code can enable the number of treatments purchased by the patient or third-party payer. The authorization code can be generated so that the code only enables treatments of the particular medical device associated with the patient for whom payment was received. For example, the authorization code may be encoded or encrypted so that only the particular medical device associated with the payment can decode or decrypt the authorization code. The authorization code may include or be transmitted with a unique device identifier, and a medical device can be configured to enter an authorization code only when a device identifier of the medical device matches the device identifier received with an authorization code.

(129) The server system can transmit the authorization code to the medical device (922).

(130) The authorization code may be transmitted, for example, over a cellular communication link to the medical device that the server system identified as being associated with the patient.

(131) The medical device can receive the authorization code (924). The authorization code may be a new authorization code that is received after an initial or prior authorization code that authorized different treatments. The new authorization code can be received after the medical device has provided an indication to the patient that more treatments need to be purchased, and the new authorization code can identify a number of additional treatments for which payment has been received.

(132) Referring to FIG. 9B, the medical device can decrypt or decode the authorization code (926). In one implementation, the medical device may decrypt the authorization code using a unique device identifier or decryption key stored on the medical device. The medical device may determine that an authorization code is authentic or intended for the particular medical device that received it. The medical device may determine, using decrypted or decoded data, what treatments are authorized. For example, the medical device may determine that a particular number of treatments are authorized. The medical device may also determine that treatment using the medical device should be modified in some way, for example, that two treatments are authorized each day instead of one treatment each day, or that the intensity of ultrasound produced should be changed.

(133) The medical device can store authorization information (928). For example, the medical device can store the number of treatments that the authorization code indicates should be authorized. The medical device can also store the authorization code received, the authorization information extracted from the received data, and other authorization information. In some instances, treatments using the medical device can be authorized without any input or action by the patient. For example, when a third-party payer sends payment to the server system, the medical device can be authorized without involvement from the patient.

(134) The medical device can determine whether treatment is authorized (929). For example, the medical device can determine whether treatment is authorized based on the authorization code that was accessed by the medical device. The medical device can determine whether treatment authorized in response to receiving user input that treatment should be provided, in (902) or through later inputs. In the situation that the patient has attempted treatment with the medical device in (902), and subsequently entered payment needed to authorize treatment in (906), the medical device can proceed to apply the treatment after the determination is made that treatment is authorized. For example, the payment and authorization process may occur quickly so that the patient perceives very little delay between entering payment and the initiation of treatment. In one implementation, treatment may begin automatically when the patient has previously attempted to initiate treatment. Treatment may alternatively be delayed until the patient imitates treatment again or confirms that treatment should proceed.

(135) If the determination indicates that treatment is authorized, the medical device can control a driver circuit to apply treatment (930). For example, the medical device may control an ultrasound transducer driver circuit in a manner that causes one or more ultrasound transducers to produce

ultrasound with therapeutic properties. For example, the driver circuit can be activated to drive one or more ultrasound transducers. The driver circuit may continue to drive the ultrasound transducers until treatment is complete. Of course, if the determination indicates that treatment is not authorized based on the authorization code (for example, if the authorization code is for a different medical device, or if the treatments authorized by that code have already been expended), the medical device can control the driver circuit so that treatment is not applied, for example, by not activating the driver circuit so that treatment is prevented.

(136) The driver circuit can be deactivated when treatment is finished (932). The medical device can store a record of the treatment applied (934). The medical device can also decrease the number of treatments authorized for the medical device (936). For example, if the medical device had received an authorization code that authorized twenty treatments of the medical device, after one treatment is completed, the medical device may update the number of authorized treatments to reflect that only nineteen treatments are currently authorized for the medical device.

(137) Referring to FIG. 10, an example of a system 1000 for collecting compliance information includes the medical device 10, a server system 1002, and a third-party system, all connected by a network 1008. The medical device 10 and the server system 1002 may, additionally or alternatively, communicate through a cellular network 1009.

(138) The medical device 10 can record use of the medical device 10, for example, by recording the date, time, and duration that treatment occurs using the medical device 10. The recorded use of the medical device 10 can be collected as compliance information 1010. For example, the medical device 10 can collect information about multiple treatments performed or multiple aspects of treatment.

(139) As described above, in some implementations, the medical device 10 may receive information about the treatment regimen for the patient. For example, the treatment regimen may be entered on the device after the health condition has been diagnosed or after the medical device 10 has been prescribed to the patient. The compliance information 1010 may, but is not required to, include one or more results of a comparison between the recorded use of the medical device 10 and the treatment regimen of the patient.

(140) To distinguish the medical device 10 from other medical devices 10, the medical device 10 may store a device identifier 1012 that enables the medical device 10 to be identified. For example, the device identifier 1012 may uniquely identify a particular medical device 10.

(141) The medical device 10 can send the compliance information 1010 to the server system 1002. Compliance information 1010 can be sent automatically, for example, after a predefined number of treatments are performed, after a particular amount of time has elapsed, or after a treatment regimen has been completed. The medical device 10 may send the compliance information 1010 over the network 1008. Additionally, or alternatively, medical device 10 may send the compliance information 1010 over a cellular network 1009. The medical device 10 can also send the device identifier 1012 to the server system 1002 with the compliance information 1010, enabling the server system 1002 to associate the compliance information 1010 with the particular device. In addition to a device identifier, or alternatively, the medical device 10 may send a patient identifier with the compliance information 1010.

(142) The server system 1002 can be configured to receive the compliance information 1010 and the device identifier 1012 from the medical device 10. The server system 1002 can determine a patient identifier using the device identifier 1012. For example, the server system 1002 may store records that associate device identifiers 1012 for multiple medical devices 10 with patient identifiers. The server system 1002 can compare the received device identifier 1012 to stored device identifiers to determine a patient identifier and identify the patient that uses the medical device 10. The server system can store the compliance information 1010 in association with the determined patient identifier.

(143) The server system 1002 can be configured to receive compliance information from multiple

medical devices **10** operated by different patients. For example, the server system **1002** may include a compliance database of many patients and information about the prescriptions and medical devices **10** that correspond to each patient. The server system **1002** can receive and record compliance information from each or any of patients or medical devices **10** described in the compliance database. The server system **1002** can add information about additional patients and medical devices **10** to the compliance database records.

(144) The server system **1002** can provide access to the stored compliance information **1010** to one or more parties that have a relationship with the patient. For example, an insurance company for the patient may use the compliance information **1010** to determine whether the patient is using the medical device **10** that the insurance company paid for. A physician or a caretaker of the patient may use compliance information **1010** to determine if the patient is complying with a treatment regimen that has been prescribed. To obtain compliance information **1010** for the patient, a third-party system **1004** may submit an inquiry requesting the compliance information **1010** to the server system **1002**. For example, the inquiry **1016** may include one or more patient identifiers to identify one or more patients. The server system **1002** can provide access to the compliance information **1010** for one or more users in response to receiving the inquiry from the third-party system **1004**.

(145) Because the server system **1002** can store compliance information about multiple patients and multiple medical devices **10**, the server system **1002** can provide aggregate information about multiple patients and medical devices **10**. For example, the server system **1002** may provide an insurance provider with a summary of the treatments performed for all patients covered by the insurance provider. As another example, a physician may receive compliance information for each of his patients from the server system **1002**, without being required to interface with each of the prescribed medical devices **10** individually.

(146) Summaries, reports, graphs, and comparisons can be provided based on compliance data for multiple patients, including, for example, information about a set or subset of patients. For example, the compliance of patients that have a particular health condition can be provided.

(147) Access to the compliance information **1010** can be limited based on the relationship of the third party to the patient. The server system **1002** can store records that associate various third parties with various patients. The third party system **1004** may be required to be authenticated or comply with other security measures before access to compliance information **1010** is provided. Access to compliance information **1010** can also be limited by restricting the quantity or detail of information available. For example, one third party may receive more detailed compliance information **1010** than a different third party may receive for the same patient. For example, an insurance company for the patient may be provided access only to the number of treatments performed with the medical device **10**, but the physician of the patient may be provided access to the particular dates and times that treatments occurred in addition to the total number of treatments.

(148) The medical device **10** can also detect and record information about errors of the medical device **10**. During treatment or during other operation of the medical device **10**, one or more errors may occur. The medical device can store information about the errors as error information **1014**, and can send the error information **1014** to the server system **1002** with the device identifier **1012**. In one implementation, error information **1014** can be sent soon after the error is detected. Examples of errors that can be detected, and for which information can be recorded and sent, include a gel error indicating that there is insufficient ultrasound conductive gel on a transducer, a battery error that indicates that remaining power of the battery is low, and a connectivity error that indicates that a wire to a transducer is disconnected or broken.

(149) The server system **1002** can be configured to receive the error information **1014** and the device identifier **1012**. The server system **1002** can store the received error information **1014** and can associate error information **1014** with the device identifier **1012**. The received error information **1014** may quickly and accurately indicate which medical devices **10** and which types of medical devices **10** experience errors and at what frequency errors occur.

(150) The received error information may also enable the server system **1002** to provide information to the medical device **10** to address the errors. For example, the server system **1002** may use error information **1014** to determine a possible cause of an error. The server system **1002** may select service information **1018** that addresses the error. For example, the server system **1002** may select information to store on the medical device **10**, which may include information to restore or replace outdated or incorrect information. The server system **1002** may select control instructions to be executed on the medical device **10**, for example, control instructions to clear an error or to reinitialize the medical device **10**. The server system **1002** can send the selected service information **1018** that addresses one or more errors to the medical device **10**. The service information **1018** can include, for example, software or firmware updates, instructions to the user of the medical device **10**, instructions to trained service personnel, and/or control instructions to alter the functioning of the medical device **10** and modules coupled to the medical device **10**.

(151) The medical device **10** can send information including one or more messages **304** to the medical device **10**. The messages **304** may be stored on the medical device **10** and displayed to the patient during treatment. The messages **304** can include updated information or additional information to add to the variety of messages **304** already stored on the medical device **10**. In addition, one or more messages **304** can be provided to instruct the patient how to correct an error of the medical device **10**, or to inform the patient that an error has been corrected.

(152) Referring to FIG. **11**, a medical device **10** may collect and send information as illustrated in the process **1100**. The processing device **50** of the medical device **10** may, for example, execute instructions stored on the storage device **52** to perform the process **1100**.

(153) A driver circuit is controlled to produce ultrasound with therapeutic properties (**1102**).

(154) Compliance information is collected (**1104**). Collecting compliance information can include recording information about use of the medical device, for example recording the number of treatments that are performed. Compliance information can include a number of treatments provided by the medical device, a date and time that a treatment was provided by the medical device, and/or a duration that a treatment was provided by the medical device. Information about multiple uses or treatments with the medical device can be collected.

(155) A treatment regimen that identifies a prescribed use of the medical device can be identified. For example, information about a treatment regimen may be entered on the medical device or received from a network, which may include a cellular network. The information about the recorded use of the medical device can be compared to the information about the prescribed use of the medical. Information indicating the degree that the recorded use matches the prescribed use can be generated.

(156) A device identifier can be accessed (**1106**). The device identifier can be stored on the medical device. The compliance information can be sent (**1108**). The accessed device identifier can be sent with the compliance information. For example, the compliance information can be sent to a server system configured to receive the compliance information. Compliance information can be sent automatically after a predetermined number of treatments have been performed. Compliance information may be sent using a wireless module, and the wireless module can include a cellular transceiver. For example, compliance information may be sent to a server system over a cellular network using the cellular transceiver. The medical device can include a SIM card that associates a particular telephone number with the medical device.

(157) One or more errors of the medical device can be detected. Information about the detected errors can be sent with the accessed device identifier to a server system. Service information to address the detected errors can be received from the server system.

(158) Referring to FIG. **12**, a server system can process information according to the illustrated process **1200**. For example, one or more storage devices can store instructions that, when executed by one or more processing devices, cause the server system to perform the process **1200**.

(159) Compliance information can be received (**1202**). For example, compliance information can

be received over a network or cellular network from one or more medical devices. Compliance information for multiple medical devices operated by different patients can be received. For example, the server system may be configured to receive compliance information from each of a plurality of patients, and each patient can be associated with at least one of a plurality of medical devices. The compliance information can include a number of treatments provided by a medical device, a date and time that a treatment was provided by the medical device, and/or a duration that a treatment was provided by the medical device. Compliance information can be received for multiple medical devices operated by different patients.

(160) A device identifier can be received (**1204**). The device identifier can be received with the compliance information. A patient identifier can be determined (**1206**). For example, the received device identifier can be used to determine the patient identifier. The server system can store records that associate device identifiers with one or more patient identifiers. The recorded device identifiers can be compared to the received device identifier to determine a recorded device identifier that matches the received device identifier.

(161) The compliance information can be stored (**1208**). The compliance information can be stored in association with the patient identifier that was determined to correspond to the compliance information. For example, the server system may be configured to store compliance information for each of a plurality of patients, and each patient can be associated with at least one of a plurality of medical devices. Compliance information for each patient can be stored in association with one or more device identifiers and/or patient identifiers.

(162) Access to the compliance information can be provided (**1210**). For example, access may be provided to one or more users of the server system. Users may submit an inquiry to the server system, and the inquiry can be associated with a patient identifier. Users can include, for example, one or more of a representative of an insurance provider of the patient, a physician of the patient, and a caretaker of the patient. Access to the compliance information for a particular patient can be provided to the users in response to receiving the inquiry that is associated with the patient identifier for the particular patient. Access to the compliance information can be limited based on the relationship of the user to the patient.

(163) For example, a physician of the patient may receive access to only a portion of the compliance information, such as only the number of times the medical device was used. A physician, on the other hand, may receive access to more detailed compliance information, such as the date, time, and duration that the medical device was used.

(164) Because compliance information for multiple patients and multiple medical devices can be stored, access to compliance information for multiple patients and multiple medical devices can be provided. For example, compliance information can be provided to a third party for treatment performed by multiple medical devices operated by multiple patients.

(165) Users that can receive access to the compliance information can include, for example, one or more caretakers, physicians, and representatives of insurance providers for any of the multiple patients whose compliance information is stored. Access can be provided to compliance information for each of a plurality of patients, where each patient is associated with at least one of a plurality of medical devices. Access to the compliance information for a particular patient can be provided to the users in response to receiving the inquiry that is associated with the patient identifier for the particular patient.

(166) Information about one or more errors of the medical device can be received. Based on the information about the errors, service information to address the errors can be selected and transmitted to the medical device. One or more messages for the patient can be transmitted to the medical device.

(167) Referring to FIG. 13, a process **1300** for storing patient information can begin with a physician writing a prescription for a patient for treatment using a medical device (**1302**). To carry out the prescribed treatment, a medical device can be dispensed to the patient (**1304**). The medical

device can be authorized at the time the medical device is dispensed or at a later time.

(168) Records for the patient and the dispensed medical device can be entered into a database **1314** (**1306**). For example, the database **1314** may store a patient record **1310** that associates a particular medical device or treatment with a particular patient, in the example, a patient named “John Smith.” The database **1314** may also store a prescription record **1312** that indicates the number of treatments that can be purchased for the patient. The number of treatments indicated in a prescription record **1312** may be authorized for application by the medical device after payment has been received for the prescribed treatments. For example, the treatments can be enabled after a third-party payer agrees to pay for the treatments or after the patient enters payment at the medical device. The database **1314** may also store other records including records that identify patient identifiers and medical device identifiers. The information in the database can be accessed by one or more client devices **1318**, **1320**, a server system **1322**, or other systems. For example, the server system **1322** may use the patient record **1310** to match payment to a particular medical device or patient. The records stored in the database **1314** may also be used to inform a third-party payer or patient the number of treatments that should be purchased to enable a treatment plan to be carried out.

(169) Referring to FIG. **14**, the authorized use of the medical device **10** can be limited to a particular patient and/or geographic area. In some instances, it is desirable by an automated method to limit the use of a medical device by an unauthorized person or in an unauthorized geographic area, such as in a country where the device has not received regulatory approval or in which an unauthorized sale of the device circumvents the chain of distribution of the device. Other examples where “geolocking” of a medical device may be desirable is where devices are sold at different price points in different regions and it is desirable to limit the ability of a lower priced device to be sold in a higher priced region, where device have region specific chargers, include specific languages, or are designed for single patient use or multiple patient use such as in some hospitals.

(170) It is also desirable to limit a single patient use device from being shared with other than the intended patient by, for example, resale or shared use with friends or family members, particularly where the device is a prescription device. The desire to limit such use is particularly applicable to medical devices such as the Exogen™ Ultrasound Bone Healing System sold by Smith & Nephew, Inc. that is easily transported.

(171) To limit, and preferably prevent, the use of a medical device **10** (FIG. **14**) in an unauthorized geographic area, the medical device **10** includes hardware and software that determine the geographical location of the medical device **10** upon start-up and compare the determined geographical location to an authorized geographical location. The authorized geographical location can be, for example, programmed in the medical device **10** during the manufacturing process.

Referring to FIG. **15**, the medical device **10** include a GPS receiver **1510** to determine geographical location, and memory **1514** to store authorized operating areas for the medical device **10**. Alternatively, the medical device **10** can use cell phone networks or Wi-Fi to determine geographical location.

(172) Referring to FIG. **16**, the location of the medical device **10** is compared to an approved area **1618** in which treatment is authorized. If the geographical location **1620** of the device **10** is within the authorized area **1618**, treatment begins. If the geographical location **1622** of the device **10** is in an unauthorized area **1624**, no treatment is delivered. The device can include two-way communication, such as cellular, internet or wireless communication. To accommodate patient travel, the medical device **10** can be configured to accept over-the-air-updates to the authorized geographical location.

(173) As illustrated in FIG. **17**, for “geolocking” purposes, the medical device **10** includes the GPS receiver **1712**, an antenna **1730** connected to the GPS receiver **1712** via an amplifier **1732** and a filter **1734**, and a crystal **1736**, an integrated circuit **1738**, and a regulator **1740**.

(174) Advantageously, a medical device can be designed for use within only a specific

geographical area, for example, a country or region as illustrated in FIG. 16, such that use of the device in unintended markets is limited or prevented.

(175) To limit, and preferably prevent, the use of the medical device **10** by an unauthorized person, the medical device **10** requires proof of patient identity using, for example, a key card or token issued to the user, a password, or physical evidence. For example, referring to FIG. 2, the medical device **10** includes hardware and software and a user interface **70** implemented on the device to identify the patient using a fingerprint, retinal scan, or voice recognition.

(176) The identity of the user is compared to the stored patient identity and treatment is only authorized when the user is confirmed to be the patient.

(177) As illustrated in FIG. 18, for device authorization purposes, the medical device **10** includes, for example, a fingerprint sensor **1852**, control electronics such as an integrated circuit **1854**, and an EPROM **1856**.

(178) Advantageously, the use of a fingerprint, retinal scan, or voice recognition ensures that the patient is at least present during the treatment. In situations where the patient is, for example, young or elderly, the identity of a supervising individual can alternatively or additionally be required. The fingerprint, voice or retinal scan of the authorized user is saved into the memory of the device when the device is prescribed and fitted by a responsible party, for example, a doctor, pharmacist or sales representative.

(179) Referring to FIG. 19, a method for limiting unauthorized use of the medical device **10** includes determining the geographic location of the medical device **10** and not allowing treatment to commence if the determined geographic location is outside an authorized geographic location, and/or includes determining user identity and not allowing treatment to commence if the determined user identity does not match an authorized identity. The medical device **10** includes means for determining the geographic location of the medical device **10** and means for determining user identity. Treatment is not allowed to commence if either the determined geographic location is outside an authorized geographic location or the determined user identity does not match an authorized identity.

(180) In use, the patient turns the device **10** on at step **1960**, the device performs a self-test of the clock and configuration at step **1962**, the patient, for example, scans his or her index finger over the fingerprint sensor **1852**, and the device confirms the patient identity at step **1964**. If the patient is the authorized user, the geographical location of the device is checked at step **1966**. If the patient is not the authorized user, an authorization alarm is delivered at step **1968** and treatment will not commence. If the patient is the authorized user and the device **10** is within its authorized geographical location treatment can be started at step **1970**. If the device **10** is not within its authorized geographical location, an authorization alarm is delivered at step **1968** and treatment will not commence.

(181) At steps **1972**, **1974** and **1976**, the device **10** monitors the time of use, allowing up to 20 minutes of treatment to be delivered before ending treatment.

(182) Referring to FIG. 20, an inventory tracking and medical device activation system **2000** can be used to control the distribution of medical devices **10**. The system **2000** includes an enabling device **2010** that communicates with the medical device **10**. The enabling device **2010** also communicates with a server **2020** over a network **2030**.

(183) Each medical device **10** can be provided to a distributor or physician in a disabled or deactivated state. For example, the medical device **10** can be provided in a state in which no treatments are authorized. Additionally, the medical device **10** can be provided in a state in which treatments cannot be purchased or authorized, until the medical device **10** is activated and thus made operative by an enabling device. Because the medical devices **10** are shipped and stored in inventory in an inoperative state, the potential for unauthorized use is very low.

(184) The enabling device **2010** includes the capability to activate medical devices **10**. For example, the enabling device **2010** can include an activation module including a wireless or wired

communication system to transmit activation information. The server **2020**, however, can limit the enabling device **2010**. For example, the server **2020** can authorize the enabling device **2010** to activate only a limited number of medical devices **10**. The total number of medical device activations that the enabling device can be performed can be limited (e.g., no more than 10 activations, until further authorization is received). Additionally, or alternatively, the number of activations that can be performed over a particular period of time can be limited (e.g., no more than 10 activations per month). As described in further detail below, the server **2020** can limit the number of medical device activations that the enabling device **2010** can perform so that, at any given time, the enabling device **2010** is authorized to perform a number of medical device activations no greater than the number of medical devices in a particular inventory **2013** of a particular sales representative **2012**. The enabling device **2010** is associated with the sales representative **2012**, and a unique identifier for the enabling device **2010** or for the representative **2012** associates the enabling device **2010** with the inventory **2013** of the representative **2012**.

(185) The server **2020** tracks the inventories of multiple representatives. The server **2020** can identify changes in inventories using information from reliable sources, for example, information that is verifiable or outside the control of the representatives. For example, the server **2020** receives information from manufacturers or distributors of medical devices about shipments of products to the representatives.

(186) Based on the inventory **2013** for the representative **2012**, the server **2020** adjusts the ability of the enabling device **2010** to activate medical devices **10**. For example, the server **2020** authorizes the enabling device **2010** to activate as many medical devices **10** as are in the official inventory **2013** for the representative **2012**. For example, when the medical devices **10** are shipped to the representative **2012**, the server **2020** receives information about the shipment **2014** from the manufacturer or distributor. In response to determining that the inventory **2013** for the representative **2012** has increased, the server **2020** transmits to the representative's enabling device **2010** an authorization code **2022** permitting a number of medical device **10** activations corresponding to the size of the shipment **2014**. If twenty medical devices **10** are shipped to the representative **2012**, the server **2020** sends an authorization code **2022** permitting the enabling device **2010** to activate up to twenty medical devices **10**.

(187) When the medical device **10** is purchased or dispensed to a patient, the sales representative **2012** can activate the medical device **10** using the enabling device **2010**. The enabling device **2010** communicates with the medical device **10** over a wired link or a wireless link, such as Bluetooth. For example, the enabling device **2010** supplies an activation code **2024** to the medical device **10** that unlocks the functionality of the medical device **10**. The enabling device **2010** can also supply an authorization code **2026** that authorizes a particular number of treatments to be performed with the medical device **10**. The activation code **2024** and the authorization code **2026** can be combined in a single message or code. After the medical device **10** receives the activation code **2024**, the medical device **10** determines whether the activation code **2024** is valid to activate the medical device **10** and/or one or more of its treatment modules. In response to determining that the activation code **2024** is valid, the medical device **10** may send an activation confirmation message (not shown) to the enabling device **2010** to indicate that activation was successful.

(188) After the enabling device **2010** activates the medical device **10**, the enabling device **2010** automatically decreases the number of activations that the enabling device **2010** can provide. For example, the number of activations permitted is decreased by one, from twenty to nineteen. The number of activations can be decreased in response to receiving the activation confirmation message from the medical device **10** so that the number is decreased only after successful activation attempts.

(189) Once the activations allowed by the authorization code **2022** are exhausted, the enabling device **2010** is restricted from activating additional medical devices **10**. The representative **2012** is thus restricted from activating medical devices **10** beyond those legitimately in the representative's

inventory **2013**. When a new shipment of medical devices **10** is sent to the representative **2012**, the server **2020** transmits a new authorization code that permits the enabling device **2010** to activate the medical devices **10** in the new shipment.

(190) In some implementations, the server **2020** can also communicate with the enabling device **2010** to reduce the number of activations allowed, for example, if the inventory **2013** of the representative **2012** decreases due to returning unused medical devices to the manufacturer. Accordingly, the number of activations that can be provided by the enabling device **2010** is maintained according to the current inventory **2013** of the representative **2012**.

(191) In some implementations, the enabling device **2010** sends a message **2028** to the server **2020** that indicates when an activation of a medical device **10** has occurred, allowing the server **2020** to monitor activations. The remaining number of activations currently allowed by the enabling device **2010** can also be included in the message **2028**.

(192) In some implementations, after the medical device **10** expends all of the treatments authorized by the authorization code **2026**, a second authorization code can be received to permit additional treatments to be performed. As described above, the second authorization code can be received in response to payment by a patient or a patient's insurance provider. The second authorization code can be received, for example, over the network **2030**, from the enabling device **2010**, or from a removable medium.

(193) By contrast, in some implementations, the medical device **10** returns to an inoperative, deactivated state after the treatments authorized by the authorization code **2026** are exhausted. Thus the medical device **10** must be activated with an activation code from the enabling device **2010** before further treatments are permitted.

(194) The medical device **10** can be configured to transmit information indicating usage of the medical device **10**, including compliance with a treatment regimen, to the enabling device **2010** over the same communication link used to receive the activation code. Thus, as part of the exchange, the medical device **10** receives a new activation code from the enabling device **2010**, and the enabling device **2010** receives usage information from the medical device **10**. The enabling device **2010** can then transmit the usage information to the server **2020**, with identifiers to identify, for example, the patient, prescription, and medical device **10** associated with the usage information.

(195) In some implementations, rather than being enabled by a transmission of an activation code from an enabling device, a medical device can be activated by a code accessed from a removable medium, such as an SD memory card.

(196) Referring to FIG. **21**, the system **2000** can also control medical device activations and treatment authorizations based on prescriptions and insurance payment authorizations. For example, the enabling device **2010** can be limited to activating a medical device when a valid prescription for the medical device is received, or when an insurance company authorizes payment for a prescribed medical device.

(197) The enabling device **2010** can be configured to require an authorization code before performing each activation. Each time a medical device is activated, a new authorization code is required. The server **2020** provides authorization codes over the network **2030**. The server **2020** can provide an authorization code allowing the activation of a single medical device based on a particular prescription for the medical device. If no prescription is submitted to the server **2020**, the server **2020** does not send an authorization code to the enabling device **2010**, and the enabling device **2010** is unable to activate medical devices.

(198) For example, after a physician issues a prescription for treatment using the medical device **10**, the prescription can be entered on the enabling device **2010**, which transmits identifying information **2115** identifying the prescription to the server **2020**. For example, the enabling device **2010** transmits a representative identifier and a prescription identifier in the identifying information **2115**. The identifying information **2115** may be additionally or alternatively transmitted by the medical device **10**, a physician's computer system, an insurance company computer system, or

other system. In some implementations, the identifying information **2115** can identify the prescription, the patient receiving the prescription, the doctor or office issuing the prescription, an insurance company for the patient, an insurance policy for the patient, the representative **2012**, the medical device **10** to be activated, and/or the enabling device **2010**, and the information can be stored by the server **2020**.

(199) The server **2020** verifies the prescription information, for example, by comparing the prescription information to other prescription records or by verifying an authentication signature in the received information. The server **2020** can also communicate with another server system **2110**, such as a server for an insurance company of the patient, to determine whether payment has been authorized for the prescription.

(200) If the prescription is determined to be valid, and/or if payment has been authorized, the server **2020** transmits an authorization code **2120** to the enabling device **2010**. The authorization code **2120** permits the enabling device **2010** to activate a single medical device **10** for the patient and prescription that were verified. The enabling device **2010** transmits an activation code **2124** to the medical device **10**. The enabling device **2010** also transmits an authorization code **2126** to the medical device **10**, permitting the medical device **10** to apply the number of treatments indicated in the prescription, or the number of treatments for which payment was authorized by the insurance company.

(201) The enabling device **2010** can transmit information to the server **2020** indicating that the activation of the medical device **10** was performed. Based on the initial identifying information **2115** and/or information received subsequent to the activation, the server **2020** can record the transaction identifying, for example, the patient, the prescription, the medical device **10**, the representative **2012**, the enabling device **2010**, and the authorization code **2120** associated with the activation.

(202) When no prescription is submitted to the server **2020**, the server **2020** does not send an authorization code to the enabling device **2010**, and the enabling device **2010** cannot activate the medical device **10**. Similarly, if the prescription information sent to the server **2020** is invalid or is not verifiable, or if payment is refused by an insurance provider, the server **2020** can withhold the authorization code, disallowing the activation of the medical device **10**.

(203) In a similar manner, the system **2000** can be used to track and/or evaluate passes for unpaid treatments. In some implementations, the enabling device **2010** can be authorized to provide a limited number of complimentary medical device activations or treatment authorizations for previously activated medical devices **10**. Each time a pass is provided, the enabling device **2010** transmits information to the server **2020** identifying, for example, the physician, representative **2012**, medical device **10**, and insurance provider associated with the pass. The information may also identify the patient and prescription associated with the pass. The server **2020** can thus maintain a database tracking the issuance of complimentary passes for different representatives **2012**, physicians, and insurance providers.

(204) In some implementations, the enabling device **2010** can be limited to providing passes when the server **2020** provides authorization for each pass individually. For example, the server **2020** can evaluate the circumstances of a particular patient and a particular prescription to determine whether the patient qualifies for complimentary or reduced cost treatment.

(205) The enabling device **2010** transmits a request to the server **2020** that indicates the circumstances of the desired pass. The request can indicate, for example, the medical condition to be treated, the identity of the patient, an associated prescription, the patient's income or other financial status, whether and to what degree treatment is covered by insurance, and what insurance provider covers the treatment. The request can also identify the physician, representative **2012**, enabling device **2010**, and medical device **10** associated with the request.

(206) The server **2020** receives and evaluates the request for the pass. For example, the server **2020** uses the information in the request to evaluate the patient's medical needs, economic circumstances,

and other circumstances to determine whether to authorize the activation and treatment authorization of a medical device for the patient. If the patient's circumstances meet the criteria for complimentary treatment, the server **2020** sends an authorization code to the enabling device **2010** permitting the enabling device **2010** to provide a pass for the particular patient. If the patient does not qualify for a pass, the server system **2020** withholds the authorization code necessary to activate a medical device or authorize additional treatments for the medical device. The server **2020** can record each request for a pass, each pass granted, and activations and treatments that occur based on the pass.

(207) The techniques described above are not limited to any particular hardware or software configuration. Rather, they may be implemented using hardware, software, or a combination of both. The methods and processes described may be implemented as computer programs that are executed on programmable computers comprising at least one processor and at least one data storage system. The programs may be implemented in a high-level programming language and may also be implemented in assembly or other lower level languages, if desired.

(208) Any such program will typically be stored on a computer-usable storage medium or device (e.g., CD-ROM, RAM, or magnetic disk). When read into the processor of the computer and executed, the instructions of the program cause the programmable computer to carry out the various operations described above.

(209) A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made. For example, the described features of presentation of messages selected for a particular patient, the collection and display of compliance information, and treatment authorization, and medical device locking based on identity or geographical location may be implemented for a single medical device. In addition, any subset of the features described can be implemented. Each of the message presentation, compliance information collection and display, treatment authorization, and medical device locking features may be implemented individually, separate from the other features described, or together in any combination.

(210) Accordingly, other implementations are within the scope of the following claims.

Claims

1. A method of operating a medical therapy system, the method comprising: by a medical therapy device that includes electronic processing circuitry: causing provision of a therapy to a wound of a patient; generating a compliance data reflecting a degree to which a duration of provision of the therapy has complied with a therapy prescription; and communicating the compliance data to one or more remote computing device; and by the one or more remote computing devices that include at least one processor: at a first time: determining that the compliance data does not satisfy a compliance threshold; and responsive to determining that the compliance data does not satisfy the compliance threshold, causing provision of an indication that the compliance data does not satisfy the compliance threshold.
2. The method of claim 1, wherein provision of the indication is performed on a user interface of the medical therapy device.
3. The method of claim 1, wherein the indication is provided to the patient via a cellular network.
4. The method of claim 1, further comprising, by the one or more remote computing devices: receiving the therapy prescription from a physician; and transmitting the therapy prescription to the medical therapy device.
5. The method of claim 4, wherein receiving the therapy prescription from the physician is responsive to the physician diagnosing the patient with a particular health condition.
6. The method of claim 4, further comprising, by the medical therapy device, activating provision of the therapy to the wound of the patient in response to receiving the therapy prescription, wherein provision of the therapy is performed according to the therapy prescription.

7. The method of claim 1, further comprising, by the one or more remote computing devices, transmitting an indication to the patient to contact a physician of the patient.
 8. The method of claim 1, wherein the compliance threshold comprises 80%.
 9. A medical therapy system comprising: a medical therapy device including: a driving circuitry configured to cause provision of a therapy to a wound of a patient; a transmission circuitry configured to transmit data to one or more remote computing devices; and an electronic processing circuitry configured to: generate a compliance data reflecting a degree to which a duration of provision of the therapy has complied with a therapy prescription; and cause the transmission circuitry to communicate the compliance data to the one or more remote computing devices; and the one or more remote computing devices including at least one processor configured to: responsive to a determination that the compliance data does not satisfy a compliance threshold, cause provision of an indication that the compliance data does not satisfy the compliance threshold.
 10. The medical therapy system of claim 9, wherein the medical therapy device further comprises a user interface, and wherein the indication is provided to the patient via the user interface.
 11. The medical therapy system of claim 9, wherein the indication is provided by the electronic processing circuitry responsive to a communication received from the one or more remote computing devices.
 12. The medical therapy system of claim 9, wherein the indication is provided to the patient via a cellular network.
 13. The medical therapy system of claim 9, wherein the one or more remote computing devices are further configured to: receive the therapy prescription from a physician; and transmit the therapy prescription to the transmission circuitry.
 14. The medical therapy system of claim 13, wherein receipt of the therapy prescription from the physician is responsive to the physician diagnosing the patient with a particular health condition.
 15. The medical therapy system of claim 13, wherein the electronic processing circuitry is further configured to activate provision of the therapy to the wound of the patient in response to receipt of the therapy prescription, and wherein provision of the therapy is performed according to the therapy prescription.
 16. The medical therapy system of claim 9, wherein the one or more remote computing devices are further configured to transmit an indication to the patient to contact a physician of the patient.
 17. The medical therapy system of claim 9, wherein the compliance threshold comprises 80%.
 18. The medical therapy system of claim 9, wherein the medical therapy device comprises an ambulatory medical device.
 19. The medical therapy system of claim 10, wherein the at least one processor of the one or more remote computing devices is further configured to: responsive to the determination that the compliance data does not satisfy the compliance threshold, cause the medical therapy device to provide the indication as a first graphical object on the user interface; and responsive to a determination that the compliance data satisfies the compliance threshold, cause the medical therapy device to provide an indication that the compliance data satisfies the compliance threshold as a second graphical object on the user interface.
 20. The method of claim 1, further comprising, by the one or more remote computing devices: at the first time: causing the medical therapy device to provide the indication as a first graphical object on a user interface of the medical therapy device; and at a second time: determine that the compliance data satisfies the compliance threshold; and responsive to determining that the compliance data satisfies the compliance threshold, causing the medical therapy device to provide an indication that the compliance data satisfies the compliance threshold as a second graphical object on the user interface.
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